WEEK 10

REAL ESTATE ANALYTICS DASHBOARD USING POWER BI

1. Introduction

The Week 10 project focuses on developing an interactive Real Estate Analytics Dashboard using Microsoft Power BI. The King County House Sales dataset was used to analyse property prices, living areas, bedrooms, bathrooms, waterfront properties, property conditions and monthly sales information.

The dashboard was designed using KPI cards, charts, map visualisation and interactive slicers to present real estate information in a clear and organised manner.

2. Objectives

1. To develop an interactive real estate analytics dashboard.
2. To analyse average and median house prices.
3. To calculate and display important property measurements.
4. To compare house prices based on bedrooms and bathrooms.
5. To analyse property prices based on waterfront and condition.
6. To visualise property information using ZIP codes.
7. To create interactive slicers for filtering dashboard data.
8. To present real estate information through meaningful visualisations.

3. Tools and Technologies Used

- **Power BI Desktop:** Used to develop the interactive dashboard.
- **Power Query Editor:** Used for data preparation and transformation.
- **DAX (Data Analysis Expressions):** Used to create calculations and measures.
- **King County House Sales Dataset:** Used as the primary data source.

4. Dataset Overview

The King County House Sales dataset contains information about residential properties and their sales.

The dataset includes the following important fields:

- ID – Property identification number.
- Date – Date of property sale.
- Price – Selling price of the property.
- Bedrooms – Number of bedrooms.
- Bathrooms – Number of bathrooms.
- Sqft Living – Living area of the property.
- Sqft Lot – Total lot area.
- Waterfront – Indicates whether the property has waterfront access.
- Condition – Condition rating of the property.
- Zipcode – Property location code.

These fields were used to create different visualisations and analyse real estate data.

5. Dashboard Layout and Design

A dashboard titled **Real Estate Analytics Dashboard** was created in Power BI. Different visual elements were arranged to make the information easy to understand.

The dashboard contains:

- KPI cards for important property measurements.
- A map visualisation for ZIP-code-based property information.
- Charts for analysing house prices.
- Interactive slicers for selecting price and living area ranges.
- Additional charts for exploring property and monthly sales information.

6. KPI Cards Creation

Five KPI cards were created to display important measurements of the real estate dataset.

6.1 Average House Price (Filtered)

This card displays the average house price based on the selected filter conditions.

Displayed value: **540.98K**

6.2 Median Price

This card displays the middle value of house prices in the dataset.

Displayed value: **450.00K**

6.3 Average Price per SqFt

This card represents the average property price per square foot.

Displayed value: **264**

6.4 Average Living Area

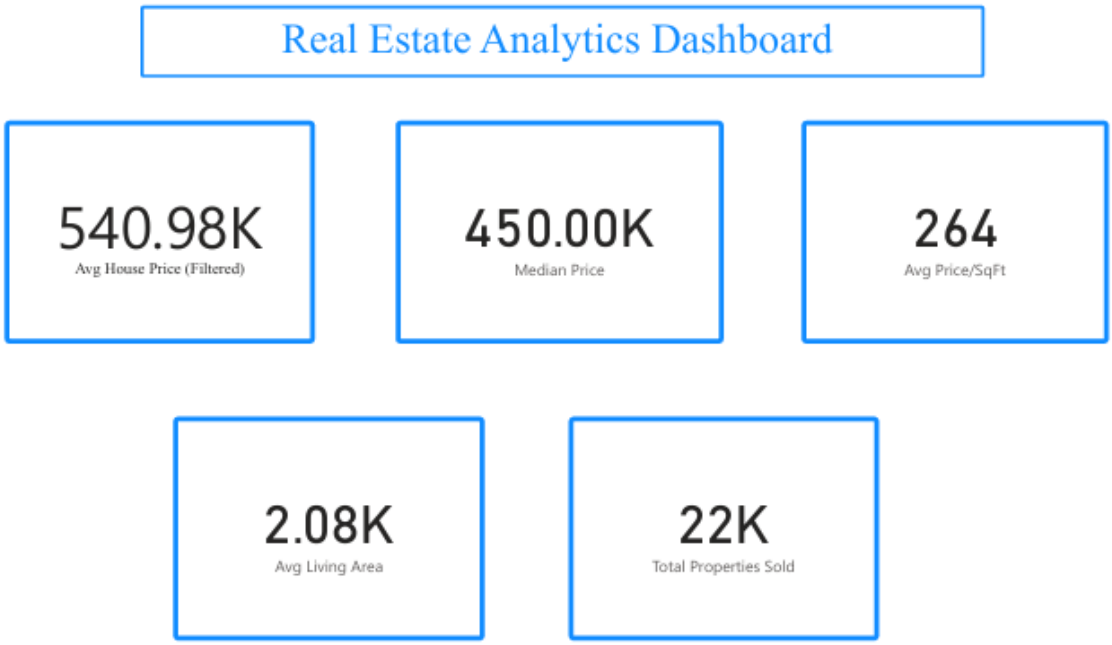
This card displays the average living area of the properties.

Displayed value: **2.08K**

6.5 Total Properties Sold

This card displays the total number of property records represented in the dashboard.

Displayed value: **22K**



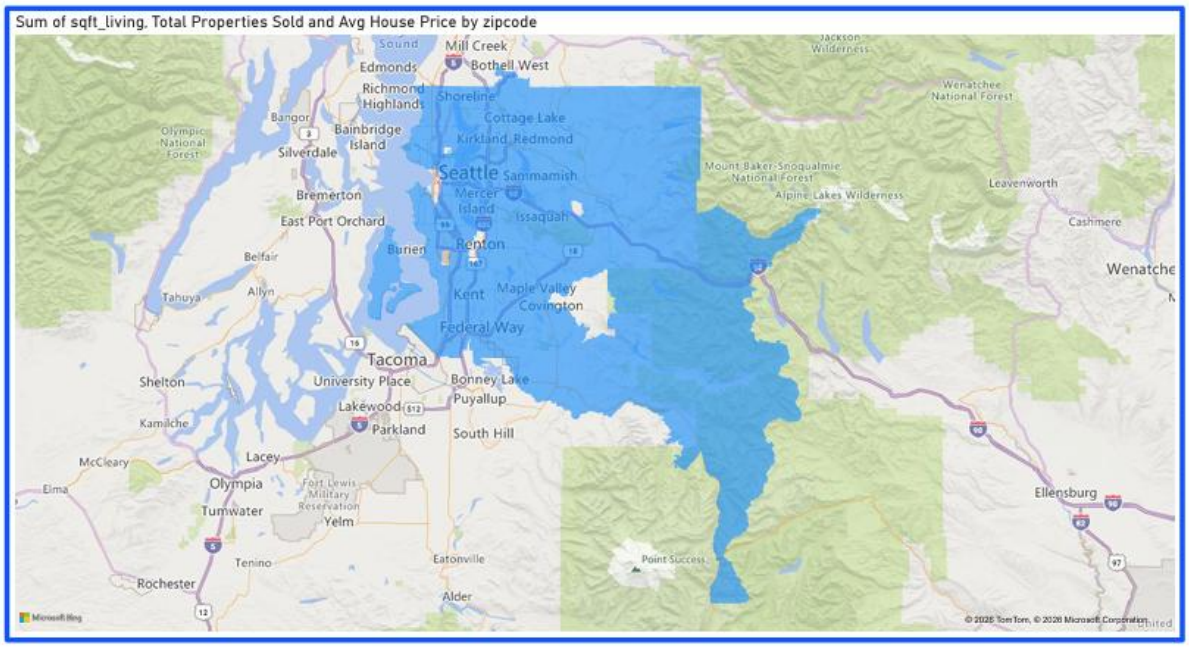
Key Performance Indicators of Real Estate Properties

7. Map Visualisation

A map visualisation was created using ZIP-code information to represent the geographical distribution of properties.

The map helps users explore property information based on different ZIP codes. It also provides a geographical view of the real estate dataset.

The visualisation includes property-related information such as living area, total properties sold and average house price by ZIP code.



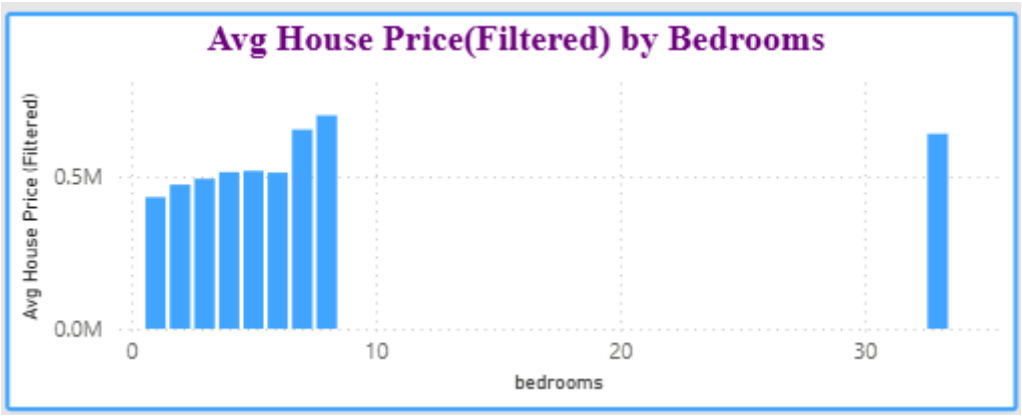
Geographical Analysis of Properties by ZIP Code

8. House Price Analysis Using Charts

Different charts were created to understand how property features relate to average house prices.

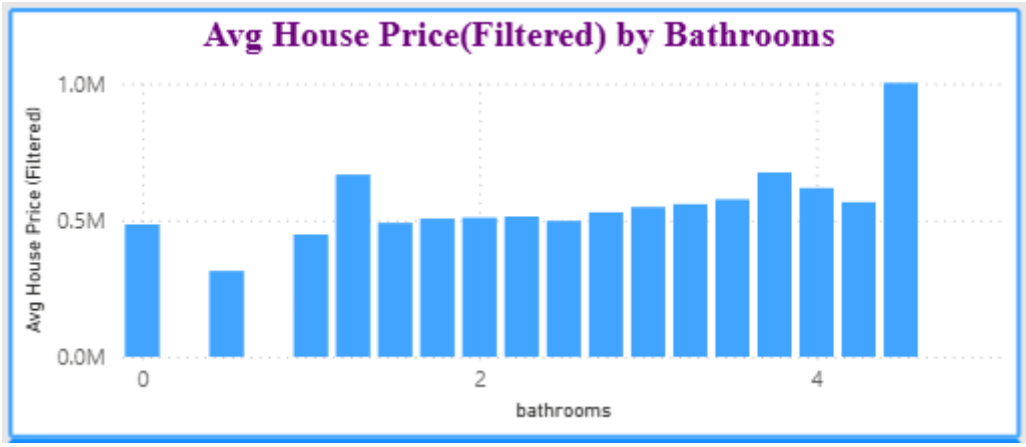
8.1 Average House Price by Bedrooms

A chart was created to compare average house prices according to the number of bedrooms. It helps users explore price differences among properties with different bedroom counts.



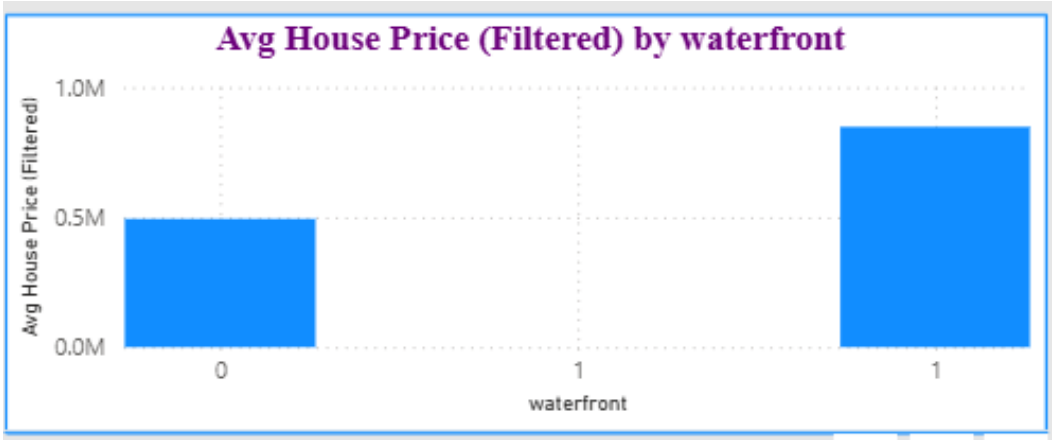
8.2 Average House Price by Bathrooms

This chart displays average house prices based on the number of bathrooms. It helps analyse the relationship between bathroom count and property prices.



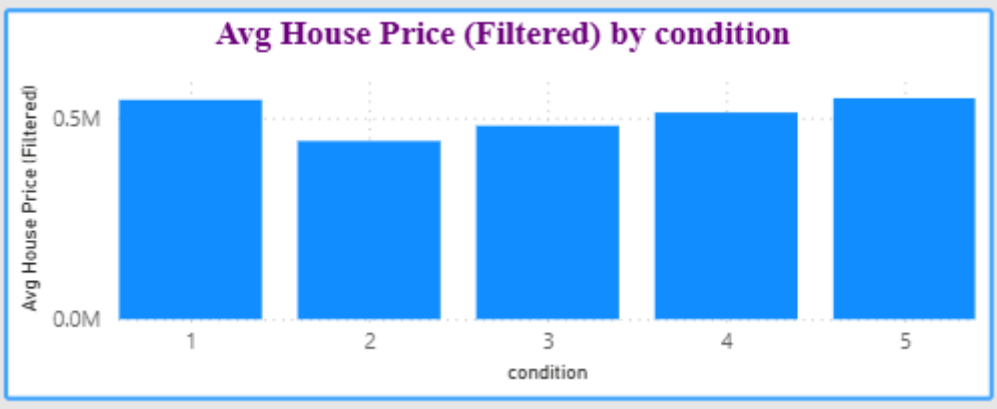
8.3 Average House Price by Waterfront

A chart was created to compare average house prices between waterfront and non-waterfront properties.



8.4 Average House Price by Condition

This chart represents average house prices according to property condition ratings from 1 to 5.



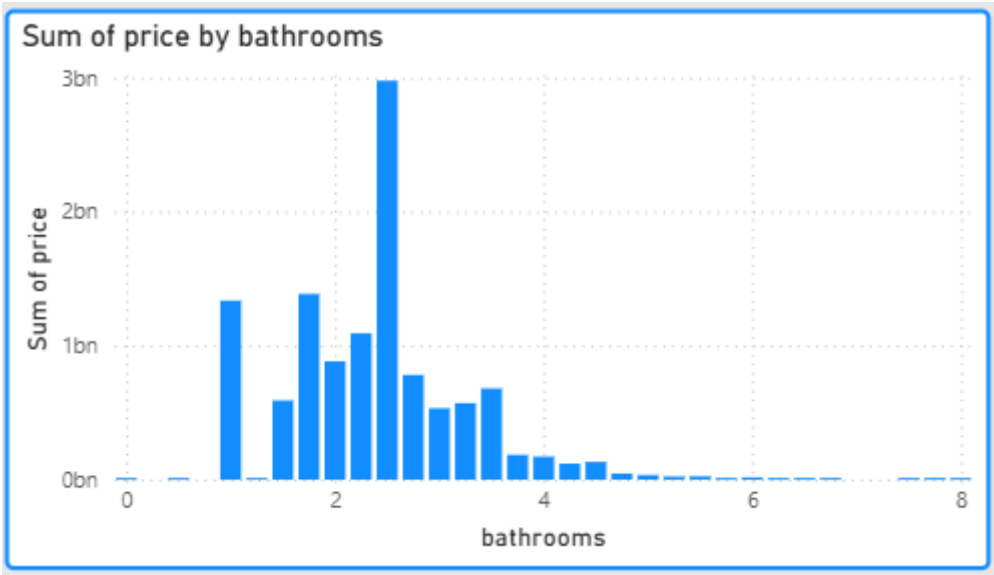
Average House Price Analysis by Property Features

9. Additional Data Visualisations

Additional charts were included to explore the relationship between property features and sales information.

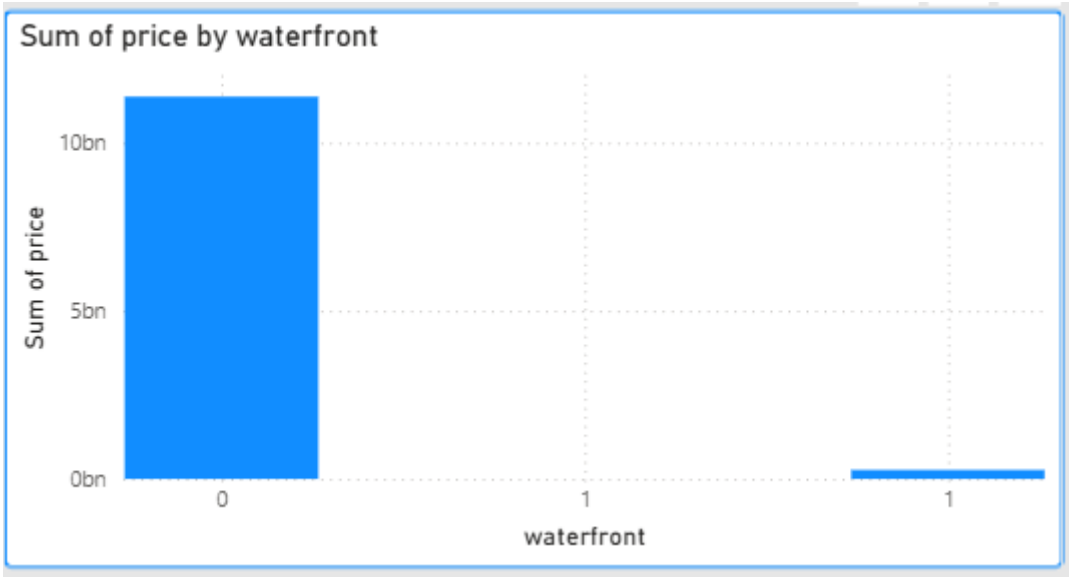
9.1 Sum of Price by Bathrooms

This chart displays the total property price according to the number of bathrooms.



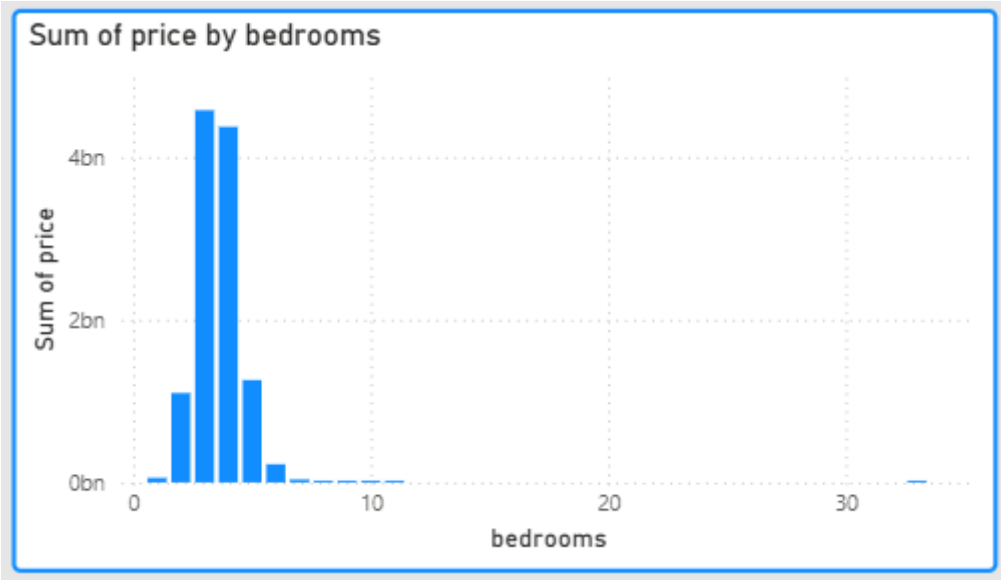
9.2 Sum of Price by Waterfront

This visualisation compares the total property price between waterfront and non-waterfront properties.



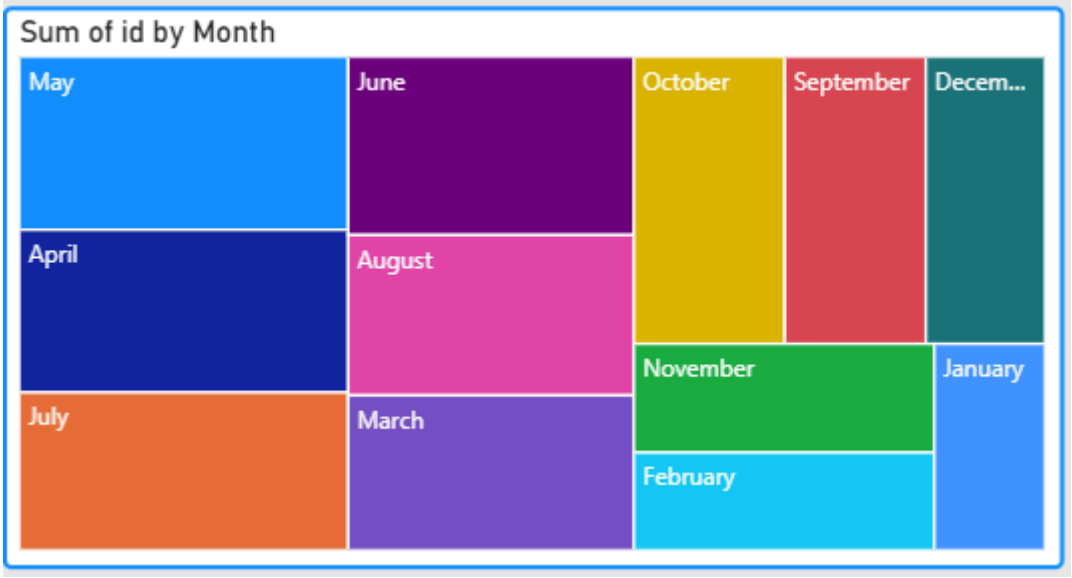
9.3 Sum of Price by Bedrooms

This chart represents the total property price based on the number of bedrooms.



9.4 Sum of ID by Month

A monthly visualisation was created using the property ID count to show the distribution of property records across different months.



Additional Property and Monthly Sales Analysis

10. Interactive Slicer Implementation

Two interactive slicers were added to make the dashboard more flexible and user-friendly.

10.1 Price Range Slicer

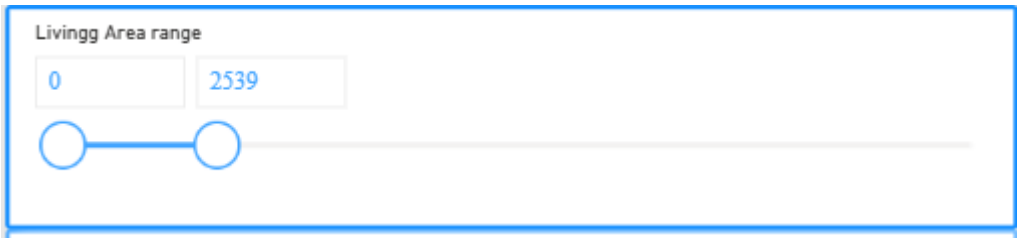
The Price Range slicer allows users to select a minimum and maximum property price. The dashboard visuals can then be explored based on the selected price range.



10.2 Living Area Range Slicer

The Living Area Range slicer allows users to filter properties according to their living area in square feet.

Both slicers help users analyse selected property categories without changing the original dataset.



Interactive Price and Living Area Slicers

11. Dashboard Interactivity

The dashboard was designed to support interactive data exploration.

- Users can select a particular price range.
- Users can select a living area range.
- KPI values and relevant charts can respond to the selected filters.
- Different charts provide comparisons across property features.
- ZIP-code information helps explore geographical property details.

This interactive functionality makes the dashboard useful for exploring different aspects of real estate data.

12. Final Dashboard Preparation

After creating the KPI cards, map visualisation, charts and slicers, all the elements were arranged on the dashboard page. The titles, labels, borders and visual sizes were adjusted to improve readability and presentation.

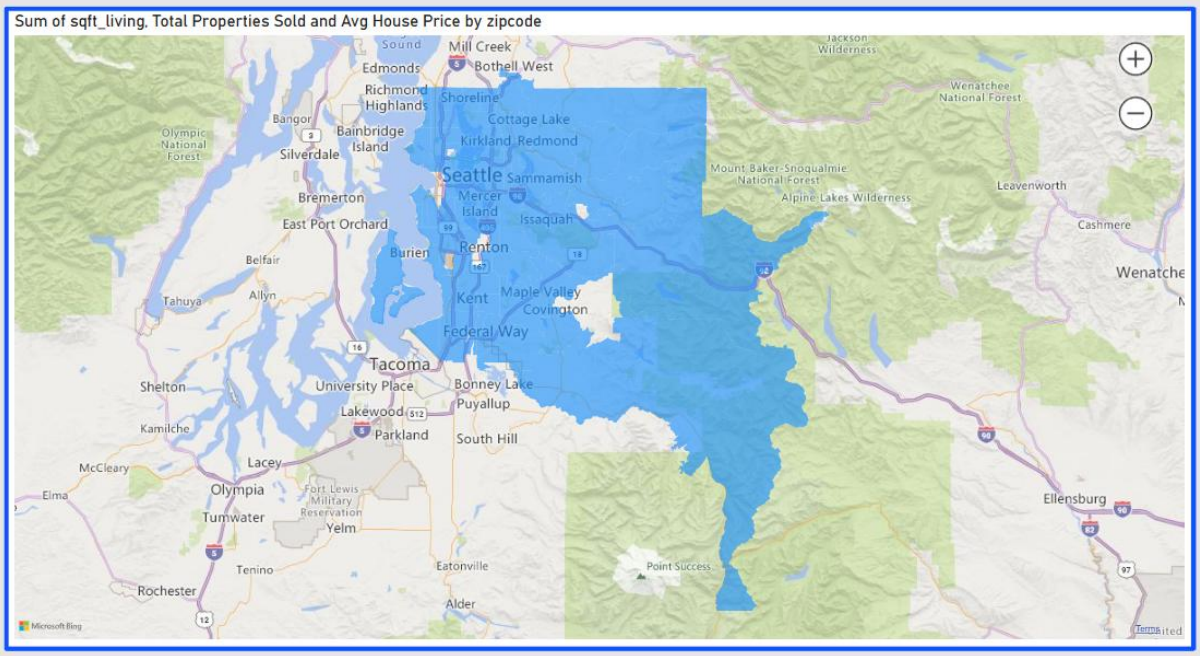
The final dashboard provides a combined view of property prices, property characteristics, geographical information and monthly records.

Divya B-E25DA013

Real Estate Analytics Dashboard

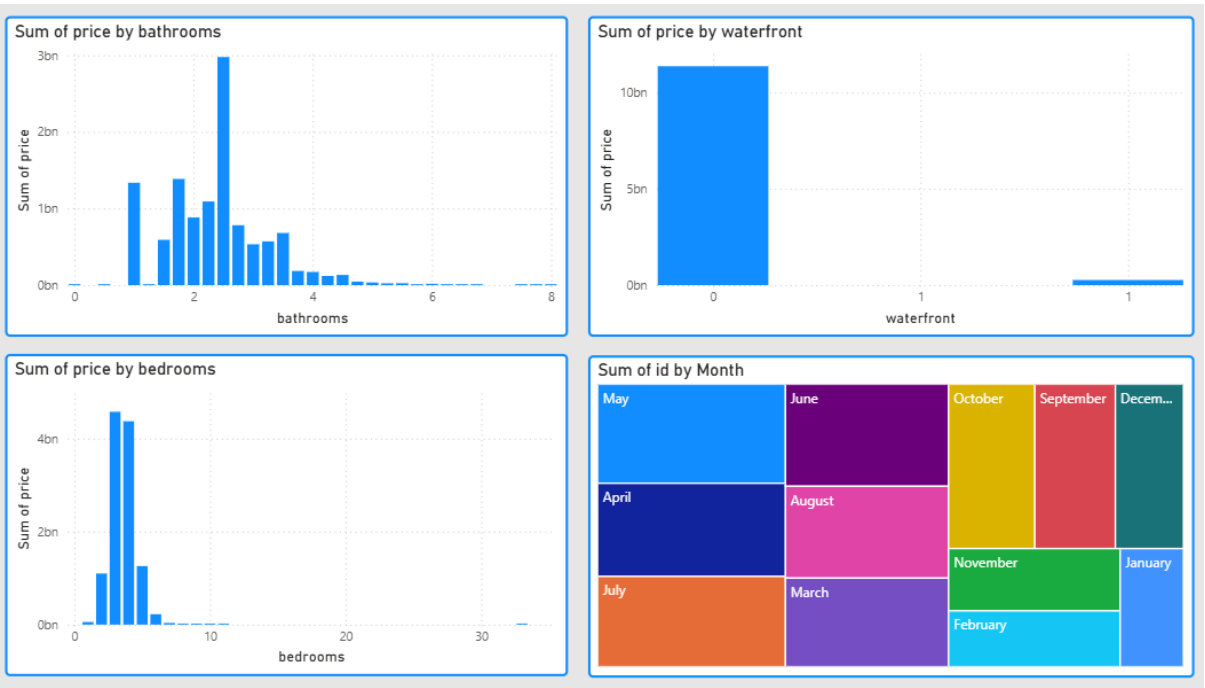
540.98K	450.00K	264
Avg House Price (Filtered)	Median Price	Avg Price/SqFt

2.08K	22K
Avg Living Area	Total Properties Sold



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Completed Real Estate Analytics Dashboard

13. Project Outcome

The project successfully transformed the real estate dataset into an interactive visual dashboard. Important property measurements were displayed through KPI cards, while charts and map visualisation provided different views of the data.

The interactive slicers enabled users to explore property information based on price and living area selections. The completed dashboard provides an organised way to examine the available real estate information.

14. Conclusion

The Week 10 Real Estate Analytics Dashboard project was successfully developed using Microsoft Power BI. Various visualisation techniques, including KPI cards, charts, map visualisation and interactive slicers, were implemented to analyse the King County House Sales dataset.

This project provided practical experience in data visualisation, DAX measures, dashboard design and interactive data analysis. The final dashboard presents real estate information in a clear, structured and user-friendly format.